What a DNS message carries

Queries and replies have the same format: a 12-byte header, then four sections.

- **Header:** an identifier copied into the reply, so the client can match answers to questions, plus a set of flags.
- **The flags that matter:** authoritative answer, recursion desired, recursion available.
- **Question:** the name being asked about, and the type asked for.
- **Answer:** the records that answer it.
- **Authority:** servers that would know.
- **Additional:** helpful records, typically the addresses of the servers just named.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer www.mit.edu A (*which is the IPv4 address of www.mit.edu?*)

```
www.mit.edu.                1800    IN      CNAME   www.mit.edu.edgekey.net.
www.mit.edu.edgekey.net.    60      IN      CNAME   e9566.dscb.akamaiedge.net.
e9566.dscb.akamaiedge.net.  20      IN      A       23.50.153.111
```

Two canonical names before an address: www.mit.edu is an alias for another name, and the machine that finally answers belongs to somebody else entirely.
Look at the TTLs: the alias is kept for minutes, the final address for seconds.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer www.mit.edu A (*which is the IPv4 address of www.mit.edu?*)

```
www.mit.edu.              1800    IN      CNAME   www.mit.edu.edgekey.net.
www.mit.edu.edgekey.net.  60      IN      CNAME   e9566.dscb.akamaiedge.net.
e9566.dscb.akamaiedge.net. 20     IN      A       23.50.153.111
```

www.mit.edu
↓ CNAME
www.mit.edu.edgekey.net
↓ CNAME
e9566.dscb.akamaiedge.net
↓ A
23.50.153.111

DNS Authority Across CNAMEs
MIT DNS: Authoritative for mit.edu, points www.mit.edu to Akamai via CNAME.
Akamai DNS: Authoritative for its own zones, resolves the CNAME chain to an IP address.

CNAMEs cross DNS zones without transferring authority over the original name.
Akamai here provides Content Delivery Network (CDN) servicies

dig — Domain Information Groper

Exercise — DNS Resolution

A user wants to visit www.university.edu and runs:

dig +noall +answer www.university.edu A

The DNS resolver returns:

www.university.edu.	*300 IN CNAME web.provider.net.*
web.provider.net.	*300 IN CNAME node42.network.net.*
node42.network.net.	*120 IN A 203.0.113.42*

Questions

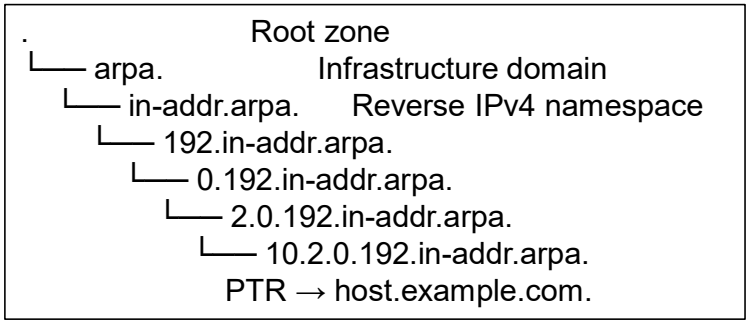
- Why does the response contain CNAME records even though an A record was requested?
- How many DNS naming hops are involved?
- Which hostname actually has the A record?
- Why does resolution move from .edu to .net?
- Assuming each domain managed by an authoritative DNS servers, which servers are responsible for each record?

dig — Domain Information Groper

From IP Addresses to Hostnames

DNS supports reverse resolution, mapping an IP address to a hostname using PTR (Pointer) records in the special DNS namespace in-addr.arpa (for IPv4), organized hierarchically through DNS zone delegations.

IP: 192.0.2.10
Reverse DNS: 10.2.0.192.in-addr.arpa.



The dig -x command performs a reverse DNS lookup:

dig -x 8.8.8.8 +noall +answer

- **-x 8.8.8.8:** Queries the PTR record associated with the IP address.
- **+noall +answer:** Displays only the answer section.

Example result: 8.8.8.8 → dns.google *8.8.8.8.in-addr.arpa. 76757 IN PTR dns.google.*

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer mit.edu MX

```
;; communications error to 10.255.255.254#53: timed out
;; communications error to 10.255.255.254#53: timed out
mit.edu.                65      IN      MX      100 mit-edu.mail.protection.outlook.com.
```

mit.edu. 65 IN MX 100 mit-edu.mail.protection.outlook.com.

↑ TTL ↑ TYPE ↑ preference ↑ mail server

MX RDATA contains two fields: a preference value and the mail server name. Lower preference values have higher priority.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

Exercise — Finding the IP Address of a Mail Server

Given the MX record:

mit.edu. 65 IN MX 100 mit-edu.mail.protection.outlook.com.

- Identify the mail server responsible for mit.edu.
- Use dig to find its IPv4 and IPv6 addresses.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

Exercise — Finding the IP Address of a Mail Server

Given the MX record:

mit.edu. 65 IN MX 100 mit-edu.mail.protection.outlook.com.

- Identify the mail server responsible for mit.edu.
- Use dig to find its IPv4 and IPv6 addresses.

- dig +noall +answer mit-edu.mail.protection.outlook.com A
- dig +noall +answer mit-edu.mail.protection.outlook.com AAAA

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer mit.edu NS

```
mit.edu.                1800    IN      NS      asia2.akam.net.
mit.edu.                1800    IN      NS      use5.akam.net.
mit.edu.                1800    IN      NS      usw2.akam.net.
mit.edu.                1800    IN      NS      use2.akam.net.
mit.edu.                1800    IN      NS      ns1-37.akam.net.
mit.edu.                1800    IN      NS      asia1.akam.net.
mit.edu.                1800    IN      NS      eur5.akam.net.
mit.edu.                1800    IN      NS      ns1-173.akam.net.
```

NAME	TTL	CLASS	TYPE	RDATA
mit.edu.	1800	IN	NS	asia2.akam.net

authoritative name server

An NS query may return multiple records: together they form an NS RRset.
Each record identifies an authoritative name server for the zone.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

Exercise — Exploring Authoritative DNS Servers

Using dig, investigate the DNS configuration of mit.edu.

- **Discover:** Find the authoritative name servers for mit.edu. How many are listed?
- **Identify:** Do the name servers belong to the mit.edu domain? What does this suggest?
- **Query:** Choose one authoritative name server and query it directly for the A record of www.mit.edu.
- **Inspect:** Does the response contain an IP address, a CNAME, or both?
- **Verify:** Repeat the query using another authoritative name server. Compare the responses.

Use dig +trace mit.edu NS to investigate how the DNS hierarchy leads to the authoritative name servers.

dig — Domain Information Groper

Exercise — Inspecting a DNS NS Query with Wireshark

Using dig, investigate the DNS configuration of mit.edu.

1. Start a Wireshark capture and apply the display filter: dns
2. Run: *dig +noall +answer mit.edu NS*
3. Locate the DNS query and its corresponding response.

Questions:

What are the source and destination IP addresses?
What are the source and destination UDP ports?
What are the QNAME (domain name being queried) and QTYPE (Type of DNS record requested)?
What is the DNS Transaction ID? Is it the same in the response?
What is the value of the QR flag (query/response) in the response?
How many NS records are returned in the Answer Section?

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer +multiline mit.edu SOA

Richiesta del record SOA (Start Of Authority) della zona

```
mit.edu.                1800 IN SOA use2.akam.net. network-request.mit.edu. (
                                23732      ; serial
                                900        ; refresh (15 minutes)
                                900        ; retry (15 minutes)
                                3600000    ; expire (5 weeks 6 days 16 hours)
                                3600       ; minimum (1 hour)
                                )
```

The SOA RDATA contains the primary name server, the responsible person's mailbox, and parameters used to synchronize secondary authoritative servers and control negative caching.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - Ask to the authoritative

dig @use2.akam.net mit.edu SOA

qr: query/response (this is a response)
aa: authoritative answer
rd: recursion desired (warning here)
ANSWER (1): requested SOA record
AUTHORITY (8): 8 authoritative name servers for mit.edu

Answer section: SOA info
MNAME: primary name server (use2.akam.net)
RNAME: responsible name (network-request.mit.edu.)
No ra: recursion not available

Authority section: NS info

```
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22041
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 8, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;mit.edu.                       IN      SOA

;; ANSWER SECTION:
mit.edu.                1800    IN      SOA     use2.akam.net. network-request.mit.edu. 23732 900 900 3600000 36
00

;; AUTHORITY SECTION:
mit.edu.                1800    IN      NS      usw2.akam.net.
mit.edu.                1800    IN      NS      use2.akam.net.
mit.edu.                1800    IN      NS      ns1-173.akam.net.
mit.edu.                1800    IN      NS      asia2.akam.net.
mit.edu.                1800    IN      NS      asia1.akam.net.
mit.edu.                1800    IN      NS      use5.akam.net.
mit.edu.                1800    IN      NS      ns1-37.akam.net.
mit.edu.                1800    IN      NS      eur5.akam.net.

;; Query time: 20 msec
;; SERVER: 96.7.49.64#53(use2.akam.net) (UDP)
;; WHEN: Sun Oct 04 19:21:11 CEST 2026
;; MSG SIZE  rcvd: 255
```

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - Ask to the authoritative

Exercise — Are All Authoritative DNS Servers Equivalent?
Run the following commands:

dig @use2.akam.net mit.edu SOA +norecurse
dig @asia1.akam.net mit.edu SOA +norecurse

Questions:

- Do both servers return an authoritative answer (aa)?
- Is the SOA MNAME the same in both responses?
- Compare the SOA serial numbers (value after RNAME). What do you observe?
- Why can two different DNS servers provide the same authoritative information?

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Ignore the local server and walk the hierarchy by hand, as it would with empty caches.

dig +norecurse @c.root-servers.net hyoka.ofc.kyushu-u.ac.jp A

- @c.root-servers.net → query directly the c root server
- hyoka.ofc.kyushu-u.ac.jp → name to resolve
- A → request its IPv4 address
- +norecurse → do not request recursion

qr
ANSWER: 0
AUTHORITY: 8
ADDITIONAL: 16 (addresses to reach the servers)

The root has no address for the host: the answer section is empty.
It delegates instead: the servers of jp in the authority section, and their addresses in the additional section

```
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53861
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 8, ADDITIONAL: 16

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 55781ae47c1fa710010000086ac28cbe9ee0105255ed1210 (good)
;; QUESTION SECTION:
;hyoka.ofc.kyushu-u.ac.jp.      IN      A

;; AUTHORITY SECTION:
jp.                     172800  IN      NS      g.dns.jp.
jp.                     172800  IN      NS      f.dns.jp.
jp.                     172800  IN      NS      c.dns.jp.
jp.                     172800  IN      NS      d.dns.jp.
jp.                     172800  IN      NS      a.dns.jp.
jp.                     172800  IN      NS      b.dns.jp.
jp.                     172800  IN      NS      e.dns.jp.
jp.                     172800  IN      NS      h.dns.jp.

;; ADDITIONAL SECTION:
a.dns.jp.               172800  IN      A       203.119.1.1
b.dns.jp.               172800  IN      A       202.12.30.131
c.dns.jp.               172800  IN      A       156.154.100.5
d.dns.jp.               172800  IN      A       210.138.175.244
e.dns.jp.               172800  IN      A       192.50.43.53
f.dns.jp.               172800  IN      A       150.100.6.8
g.dns.jp.               172800  IN      A       203.119.40.1
h.dns.jp.               172800  IN      A       161.232.72.25
a.dns.jp.               172800  IN      AAAA    2001:dc4::1
b.dns.jp.               172800  IN      AAAA    2001:dc2::1
c.dns.jp.               172800  IN      AAAA    2001:502:ad09::5
d.dns.jp.               172800  IN      AAAA    2001:240::53
e.dns.jp.               172800  IN      AAAA    2001:200:c000::35
f.dns.jp.               172800  IN      AAAA    2001:2f8:0:100::153
h.dns.jp.               172800  IN      AAAA    2a01:8840:1bc::25

;; Query time: 48 msec
;; SERVER: 192.33.4.12#53(c.root-servers.net) (UDP)
;; WHEN: Sun Oct 04 19:28:38 CEST 2026
;; MSG SIZE  rcvd: 537
```

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

$ dig +norecurse @b.dns.jp hyoka.ofc.kyushu-u.ac.jp A

;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
kyushu-u.ac.jp. 86400 IN NS ns2.kyushu-u.ac.jp.
ns2.kyushu-u.ac.jp. 86400 IN A 133.5.1.9

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

```
$ dig +norecurse @b.dns.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
kyushu-u.ac.jp. 86400 IN NS ns2.kyushu-u.ac.jp.
ns2.kyushu-u.ac.jp. 86400 IN A 133.5.1.9

$ dig +norecurse @ns2.kyushu-u.ac.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 2, ADDITIONAL: 2
ofc.kyushu-u.ac.jp. 3600 IN NS ns.jimu.kyushu-u.ac.jp.
ns.jimu.kyushu-u.ac.jp. 3600 IN A 133.5.40.1
```

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

```
$ dig +norecurse @b.dns.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
kyushu-u.ac.jp. 86400 IN NS ns2.kyushu-u.ac.jp.
ns2.kyushu-u.ac.jp. 86400 IN A 133.5.1.9

$ dig +norecurse @ns2.kyushu-u.ac.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 2, ADDITIONAL: 2
ofc.kyushu-u.ac.jp. 3600 IN NS ns.jimu.kyushu-u.ac.jp.
ns.jimu.kyushu-u.ac.jp. 3600 IN A 133.5.40.1

$ dig +norecurse @ns.jimu.kyushu-u.ac.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr aa; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
hyoka.ofc.kyushu-u.ac.jp. 3600 IN A 133.5.40.197
```

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

```
$ dig +norecurse @b.dns.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
kyushu-u.ac.jp. 86400 IN NS ns2.kyushu-u.ac.jp.
ns2.kyushu-u.ac.jp. 86400 IN A 133.5.1.9

$ dig +norecurse @ns2.kyushu-u.ac.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 2, ADDITIONAL: 2
ofc.kyushu-u.ac.jp. 3600 IN NS ns.jimu.kyushu-u.ac.jp.
ns.jimu.kyushu-u.ac.jp. 3600 IN A 133.5.40.1

$ dig +norecurse @ns.jimu.kyushu-u.ac.jp hyoka.ofc.kyushu-u.ac.jp A
;; flags: qr aa; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
hyoka.ofc.kyushu-u.ac.jp. 3600 IN A 133.5.40.197
```

root → jp → kyushu-u.ac.jp → ofc.kyushu-u.ac.jp:
at every step the same question, and a referral to the next server. Only the last reply carries the aa flag: it comes from the zone itself. dig +trace does the whole walk in one command.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

$ dig +trace +nodnssec hyoka.ofc.kyushu-u.ac.jp A

+nodnssec: Omits DNSSEC-related data (cryptographic security information) to simplify the output.

dig +trace — Tracing DNS Resolution

Shows the step-by-step DNS resolution path:

Root servers (.) → Referral to TLD name servers (.jp).
TLD servers (.jp) → Referral to authoritative servers for kyushu-u.ac.jp.
Authoritative server → Returns the final A record (IP address).

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

$ dig +trace +nodnssec hyoka.ofc.kyushu-u.ac.jp A

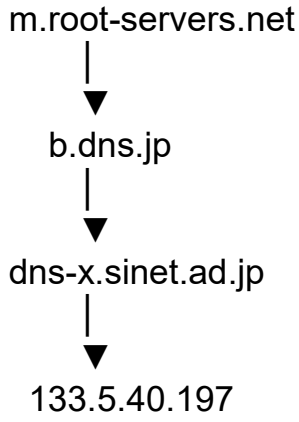
+nodnssec: Omits DNSSEC-related data (cryptographic security information) to simplify the output.

dig +trace — Tracing DNS Resolution

Shows the step-by-step DNS resolution path:

Root servers (.) → Referral to TLD name servers (.jp).
TLD servers (.jp) → Referral to authoritative servers for kyushu-u.ac.jp.
Authoritative server → Returns the final A record (IP address).

```
jp. 172800 IN NS e.dns.jp.
jp. 172800 IN NS f.dns.jp.
...
;; Received 513 bytes from 202.12.27.33#53(m.root-servers.net) in 41 ms
…
kyushu-u.ac.jp.        86400   IN      NS      dns-x.sinet.ad.jp.
;; Received 178 bytes from 202.12.30.131#53(b.dns.jp) in 312 ms
```



DNS Aliasing with CNAME Records

- DNS provides an **aliasing** service for servers in which complicated/long names (aka canonical) are associated to a more simple/short ones (aka alias).
  - **Stable names:** Applications use a consistent hostname even when the underlying infrastructure changes.
  - **Flexibility:** The canonical hostname can change without affecting the public alias.
  - **Separation of responsibilities:** The alias and its target can belong to different DNS zones managed independently. The domain owner controls the public alias, while the service provider manages the target hostname and its IP address.

Example:
www.enterprise.com
↓ CNAME
server.provider.net
↓ A
192.0.2.10

Aliases decouple public service names from infrastructure hostnames.

DNS: Load Distribution

- A DNS can be used to perform **load distribution** among replicated servers (aka, **round-robin DNS**).

- Typically, busy sites (e.g., google, amazon, CNN, etc.) are **replicated over multiple servers**, with each server running on a different end system and each having a different IP address (multiple IP addresses associated with one canonical hostname).

- The DNS database contains this set of IP addresses. When clients make a DNS query for a name mapped to a set of addresses, the server may **rotate the order in which addresses are replied**.

DNS: Multiple NS vs. Multiple A Records

Multiple NS Records — DNS Service Availability

- A DNS zone has multiple authoritative name servers.
- Resolvers can query different authoritative servers.
- Purpose: Distribute DNS queries and improve reliability.

```
example.com.  IN NS  ns1.provider.net.
example.com.  IN NS  ns2.provider.net.
example.com.  IN NS  ns3.provider.net.
```

Multiple A Records — Application Load Distribution

- A single hostname maps to multiple server IP addresses.
- DNS may rotate the order of returned IP addresses (round-robin).
- Purpose: Distribute application traffic across replicated servers.

```
www.example.com. IN A 192.0.2.10
www.example.com. IN A 192.0.2.20
www.example.com. IN A 192.0.2.30
```

Key difference: Multiple NS records distribute DNS queries; multiple A records can distribute application traffic.

Exercise

Pick a domain — `vatican.va`, `unina.it`, one of your own — and ask it five things.

```
$ dig vatican.va                    1.  the address
$ dig vatican.va NS                 2.  who has authority
$ dig vatican.va SOA                3.  which of them is the main one
$ dig a.nic.va A                    4.  its address
$ dig @212.77.0.110 vatican.va A    5.  ask it directly
```

Steps 4 and 5 use what step 3 returned: the name of the main server, then the address you just resolved for it.

Exercise

(a) Which of the five answers come without the aa flag, and why is that the expected outcome for the first three?

(b) Does the primary server named in the `SOA` record appear among the `NS` records? Should it?

(c) Which TTL comes back with the first query, and what does it drop to if you repeat the query a minute later?

(d) Step 5 asks the server that holds the zone. Is the answer authoritative? If it is not, who answered?

(e) Try an `MX` query on the same domain. How many queries does it take to reach an address you could open a connection to?

Summary

- Names are for people and addresses are for the network: the DNS is the translation, and a distributed database.
- The name space is a tree; a zone is the part of it one administration actually serves.
- Records are typed — A, NS, CNAME, MX, SOA — and each carries a TTL.
- Your host asks once, recursively; the local server walks the hierarchy iteratively, one referral at a time.
- Caching is what makes this affordable, and the TTL is what bounds how long a stale copy may keep answering.
- What dig prints — flags, answer, authority and additional sections — is the message format.